Lecture Scribe Transformation

L11_S2_A.pdf
PDF

I want a lecture scribe to refer for an exam. Here attached is the lecture notes for which I want you to give me exam ready scribe. The scribe should function as exam reference material and include proofs, derivations, and worked examples exactly as presented in the lecture.

Lecture Scribe Lesson 11
Course:
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Key results

Above written is an example Scribe. Using the same structure, generate a lecture scribe for the provided context.

Here is your **exam-ready Lecture Scribe** based strictly on the provided lecture notes and the handwritten derivations and examples shown in the slides.

Lecture Scribe Lesson 11

Course: CSE400 – Fundamentals of Probability in Computing
Lecture number: 11
Topic: Transformation of Random Variables
Date: February 10, 2026

Definitions (Key Definitions)

Definition 1: Transformation of a Random Variable

Let X be a continuous random variable with known PDF $f_X(x)$. If a new random variable is defined as

$$Y = g(X)$$

then Y is called a transformed random variable.

Objective:
Find the PDF of the new random variable:

$$f_Y(y)$$

Assumption (from lecture):

- The PDF of X, $f_X(x)$, is known a priori.

Definition 2: Cumulative Distribution Function (CDF)

$$F_Y(y) = P(Y \leqslant y)$$

CDF is used as an intermediate step to derive the PDF of transformed variables.

Definition 3: Monotonic Function

A function $g(x)$ is monotonic if:

- Strictly increasing, OR
- Strictly decreasing

This ensures inverse function exists:

$$x = g^{-1}(y)$$

Definition 4: Transformation of Two Random Variables

If

$$Z = g(X, Y)$$

then we use joint PDF:

$$f_{XY}(x, y)$$

to derive

$$f_Z(z)$$

Theorems / Propositions (Statements)

Theorem 1: PDF of Transformed Random Variable (Monotonic Increasing Case)

If:

$$Y = g(X)$$

and $g(x)$ is strictly increasing, then

$$F_Y(y) = F_X(g^{-1}(y))$$

and PDF is

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|}$$

Theorem 2: PDF of Transformed RV (Monotonic Decreasing Case)

If $g(x)$ is strictly decreasing:

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

and PDF is

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|}$$

(Same final formula applies using absolute value.)

Theorem 3: Sum of Two Random Variables

If:

$$Z = X + Y$$

then

$$F_Z(z) = P(X + Y \leqslant z)$$

and PDF is obtained using joint PDF integration.

Proofs (Main Proof Ideas / Steps)

Proof of Transformation Formula (Increasing Case)

Given:

$$Y = g(X)$$

Step 1: Write CDF

$$F_Y(y) = P(Y \leqslant y)$$

Substitute transformation:

$$= P(g(X) \leqslant y)$$

Since monotonic increasing:

$$= P(X \leqslant g^{-1}(y))$$

Thus:

$$F_Y(y) = F_X(g^{-1}(y))$$

Step 2: Differentiate to obtain PDF

.

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using chain rule:

$$= f_X(g^{-1}(y)) \frac{d}{dy}(g^{-1}(y))$$

Thus:

$$\boxed{f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|}$$

evaluated at

$$x = g^{-1}(y)$$

Proof for Decreasing Case

Start:

$$F_Y(y) = P(Y \leqslant y)$$

Since decreasing function:

$$= P(X \geqslant g^{-1}(y))$$

Thus:

$$= 1 - F_X(g^{-1}(y))$$

Differentiate:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$$

Derivations / Equations (Important Derivations)

General Formula for Transformation

$$\boxed{f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|}$$

where

$$x = g^{-1}(y)$$

Derivation for Sum of Two Random Variables

Given:

$$Z = X + Y$$

CDF:

$$F_Z(z) = P(Z \leqslant z)$$

Substitute:

$$= P(X + Y \leqslant z)$$

Using joint PDF:

$$= \int \int_{x+y \leqslant z} f_{XY}(x, y)dxdy$$

This defines PDF of sum.

Examples

Example 1: Transformation Using Sine Function

(from page 5 illustration)

Given:

$$X \sim Uniform(-1, 1)$$

PDF:

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & otherwise \end{cases}$$

Transformation:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Find:

$$f_Y(y)$$

Solution:

Step 1: Find inverse transformation

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Take inverse sine:

$$\sin^{-1}(y) = \frac{\pi X}{2}$$

Solve:

$$\boxed{X = \frac{2}{\pi}\sin^{-1}(y)}$$

Step 2: Differentiate

$$\frac{dx}{dy} = \frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

Step 3: Apply transformation formula

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|$$

Substitute:

$$= \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Final answer:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

Example 2: Sum of Two Random Variables

Given:

$$Z = X + Y$$

Find:

1. PDF of Z
2. If X,Y independent
3. If X,Y normal
4. If X,Y exponential

Solution:

CDF:

$$F_Z(z) = P(Z \leqslant z)$$

$$= P(X + Y \leqslant z)$$

Using joint PDF:

$$= \int \int_{x+y \leqslant z} f_{XY}(x, y) dx dy$$

Graphically illustrated in lecture (page 7 region diagram)

Special Case: Normal Distribution

Given:

$$X \sim N(0, 1)$$

$$Y \sim N(0, 1)$$

Independent

Then:

$$Z = X + Y$$

Result:

$$\boxed{Z \sim N(0, 2)}$$

Mean:

$$\mu_Z = 0 + 0 = 0$$

Variance:

$$\sigma_Z^2 = 1 + 1 = 2$$

Remarks

1. CDF method is fundamental for transformation.
2. Absolute derivative is required.
3. Inverse function must exist.
4. Region of integration must be carefully defined.
5. Transformation preserves probability.

Assumptions

1. Original PDF known.
2. Function monotonic.
3. Differentiable inverse exists.
4. Joint PDF known for two variables.
5. Independence simplifies calculation.

Special Cases

Case 1: Increasing function

$$F_Y(y) = F_X(g^{-1}(y))$$

Case 2: Decreasing function

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

Case 3: Two variables sum

$$F_Z(z) = \int \int_{x+y \leqslant z} f_{XY}(x, y) dx dy$$

Summary

This lecture covered transformation techniques to find PDF of new random variables derived from existing ones.

Steps:

1. Start from CDF definition
2. Substitute transformation
3. Use inverse function
4. Differentiate
5. Apply derivative magnitude

Also covered transformation involving two random variables and sum distribution.

Key Results (Exam-Critical)

Result 1: Transformation Formula

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|}$$

Result 2: CDF relationship

Increasing case:

$$F_Y(y) = F_X(g^{-1}(y))$$

Decreasing case:

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

Result 3: Sum of RVs

$$F_Z(z) = P(X + Y \leqslant z)$$

Result 4: Sum of Normal RVs

$$\boxed{N(0,1) + N(0,1) = N(0,2)}$$

If you want, I can also convert this into a **clean exam-ready PDF formatted scribe with boxed formulas and solved examples exactly in exam style.